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Designing with Drafts and Cross-Sections: Reflections on How Representations of Woven Structure Shape Design Practice

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The draft, a grid of black and white squares used to specify where a particular warp crosses over or under a weft in woven cloth, is a common representation used to design and describe woven structures. The squares in a draft directly, or with a few modifications, translate to manufacturing instructions, which makes the draft a common representation used in weaving design software. In this paper, we describe how using a different woven structure representation, a cross-section, in a design tool that foregrounds different concerns. Specifically, we describe and reflect on our motivation, design, and initial testing of an interactive cross-section design tool and share how it helped us appreciate how a cross-section representation gives "voice" to the material conditions of weaving at design time — color interaction, layer structure, and loom constraints — that the draft tends to leave less immediately legible. We share our experiences to provide inspiration for future design tools and to help those interested in documentation and sharing of their designs consider how representations shape attention to different facets of the weaving process.

Keywords: weaving; cross-section; design tools; representation

1. Introduction

In weaving's vast history, there have been several modes of communicating woven structure between weavers, communities and looms. Formats include song, choreography, side-by-side demonstration, and various diagrammatic representations. The form representations take, and encodings of knowledge into these representations, shape thought and creative experimentation (Harlizius-Klück, 2017; Verbeek, 2011). They also shape how design knowledge can be accessed, used, and adapted into other contexts (Devendorf et al., 2022; Gaver, 2011; Goveia da Rocha et al., 2022; Meiklejohn et

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al., 2024). Today, one of the most common representations used to design and communicate a woven structure is the draft, which represents both the structure of the cloth (e.g. where warps travel over and under wefts) and the movements of a loom/weaver at a particular point in the production of cloth. This "production-centric" representation formed the basis of AdaCAD, an open-source tool for drafting that we have developed and maintained over the past 8 years (Devendorf et al., 2023; Friske et al., 2019). And in this time, we have come to see the specific opportunities and challenges that the draft imposes as the foundational representation of a computational design system. Namely, our use of the draft in AdaCAD lends itself to the binary logics of computation — the draft supports computational encoding and manipulation rather than being computational in itself — which opens new potentials for encoding and composing drafts but also presents the draft square as the primary unit of cloth composition. This has the unintended side effect, especially for novice weavers, of placing the focus on the aesthetics of a draft, rather than the cloth the draft will eventually produce.

In this paper, we describe how an interactive cross-section feature that we added to our software helped us appreciate how the cross-section specific representation gives “voice” to the material considerations at design time by foregrounding: color and relative yarn size; structures as time-sequenced manipulations; and detailed attention to the relationships between interlacements within complex structures. We use our first-person experiences to describe and analyze these qualities with a particular focus on how different representations aided different elements of a single design project. Our reflections have implications for future software design, but also pedagogical implications for how woven structures can be designed, communicated, manipulated, and translated across multiple loom configurations.

2. Background

2.1. Representing Woven Cloth: Draft and Cross-Section

The design and analysis of woven cloth, particularly within Western weaving traditions, relies on representational approaches that abstract from a cloth sample into transmissible forms. Two such representations are the draft and the cross-section. In this paper, we use the term draft to refer to a grid-based representation of cloth structure — a matrix of black and white square cells indicating whether the warp or weft is visible at each intersection — rather than the loom-specific configuration of threading, tie-up, and treadling from which such a grid can be derived. For those versed in shaft loom weaving, we use the term draft to describe the *drawdown*, and even more specifically, a repeating design unit within the drawdown that represents a particular woven structure. A cross-section depicts the physical interlacement of warp and weft, showing how threads pass over and under one another in a slice through the cloth. Cross-sections can be warp or weft oriented, and in our project, we focused on cross-sections created by slicing across the warps to reveal the continuous movements of multiple wefts.

These are not the only two representations used to communicate woven structure, as weavers and textile designers work with a range of notation forms, many of which overlap or exist at different levels of abstraction (Dietz, 1949; Harlizius-Klück, 2017; Strange Loop Conference, 2018).

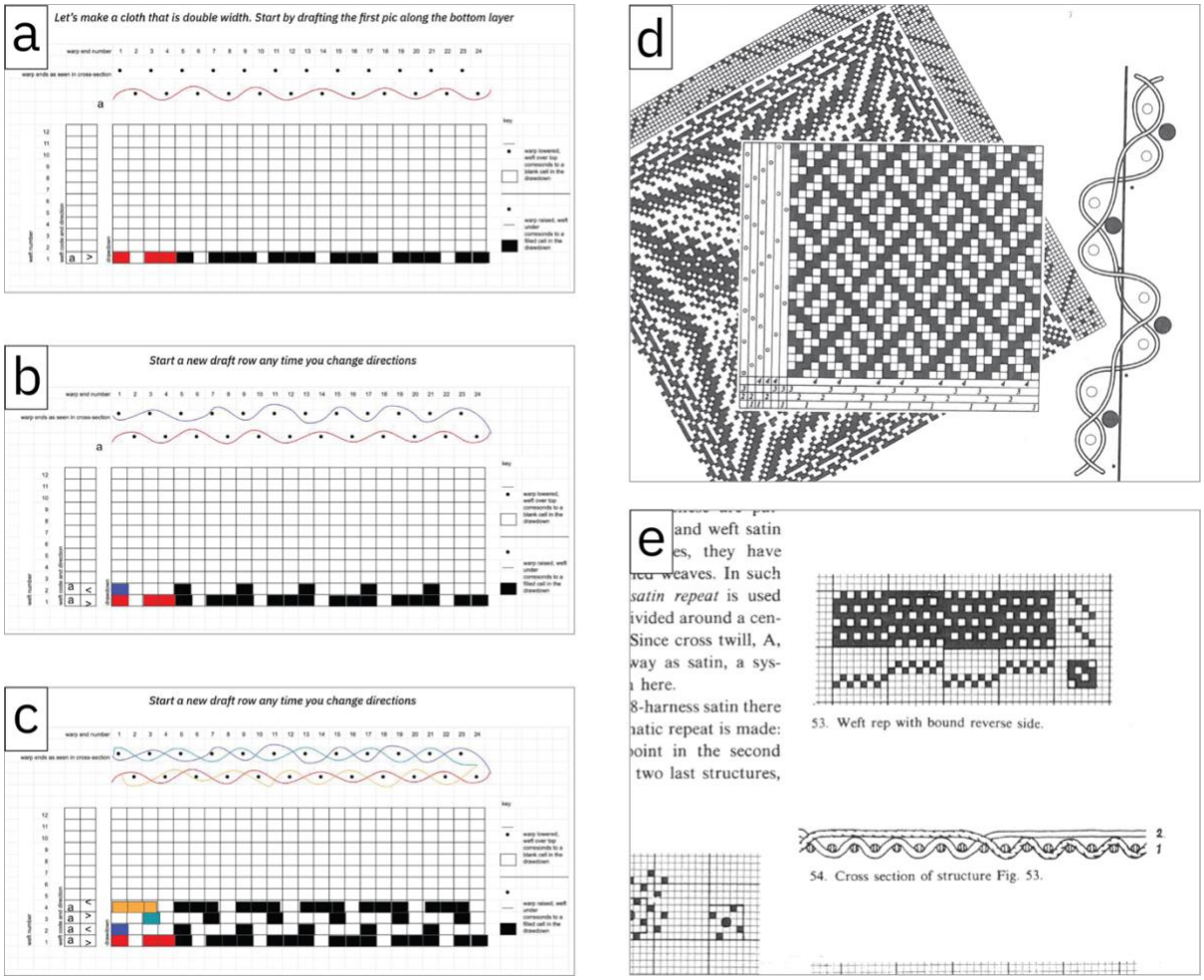


Figure 1: Examples of drafts and cross-sections from our teaching materials and weaving textbooks. (a-c): A slide sequence used in Experimental Textiles to describe how a cross-section translates to a draft. Excerpts from texts that use drafts and cross-sections side by side to explain structure, including (d) Madelyn van der Hoogt's "A Complete Book of Drafting for Handweavers" and (e) an insert from Ulla Cyrus-Zetterström's "Manual of Swedish Handweaving."

We focus on draft and cross-section specifically because they emphasize two contrasting orientations toward woven design: one leaning procedural and machine-directed, the other structural and human-interpretive. We treat these as complementary modes that foreground different concerns, not as mutually exclusive categories — each can, with effort, express what the other makes immediately legible.

2.2 Weaving Software and its Representations

Philosopher of Technology Peter-Paul Verbeek provides language for describing how a design interface mediates a design process. Specifically, the representations used by the design system make various material-semiotic qualities of the design perceptible through what their representations make present and actionable (Verbeek, 2006, 2011). Weaving tools do not *determine* what a weaver will do, but the representations they use make some connections, procedures, imaginaries more readily available than others. We find the term "voice" to be a useful way of describing this dynamic: a way to enact notions of human and material agencies in a design "conversation" (Ingold, 2013; Schon, 1984). A design tool participates in this conversation almost like a moderator, structuring what voices — the yarn's behavior, the loom's constraints, the structure's logic — can be heard and responded to through the process.

Currently, most weaving software operates as a translator between design intent and machine operation — the structure of the machinery largely influences the software design. For floor looms, programs like WeavePoint and FiberWorks locate manipulation in the loom threading, tie-up and treadling: emphasizing the key role of machine configuration in crafting a domain for exploration on the loom. Multi-shaft dobby and Jacquard-oriented software like PointCarre, EAT, ScotWeave and ArahWeave tend to prioritize the "drawdown", readily offering comparisons between the draft representation the imagery, often offering designs a library of structures with which they can "fill" in graphic regions of the design. Due to barriers to accessing Jacquard software, many weavers use Photoshop to compile drafts, further linking draft making with image manipulation (Schlein & Ziek, 2006). In most of these tools, alternative views — simulations, yarn paths, color previews — serve to check a draft rather than to produce one.

Artists and researchers are always pushing the boundaries of design, using representations that blend weaving with synthesizer representations (*Wyrd Drafts*, n.d.); AR or 3D simulation driven design approaches (*Augmented Weaving*, n.d.; Royal College of Art, London, United Kingdom & Roth, 2023; Voorwinden & McQuillan, 2025); and/or bespoke programs that convert text, data, or another input signal into a woven pattern (Hayama & Uchibe, 2025). WeaveCraft, a research prototype created by the Virtual Textiles Group at RISD (Wu et al., 2020), is the first system we have seen that uses a cross-section as the primary design representation. Their representation was specifically used to make complex 3D structures, emphasizing the use of multiple warp and weft systems as layers that interlock and relate in three-dimensional configurations. WeaveCraft inspired us to consider the cross-section as a design representation more seriously, and in this project, we reflect on our design and use of a cross-section first design tool to understand what interactive cross-section design affords in terms of action and perception.

3. Collaborative Design, Use, and Reflection on Cross-Section Design

The cross-section design feature is a collaboration between this paper's authors. Gelosi designed and implemented the system and Devendorf served as a tester of the feature. Devendorf is a weaver and design researcher trained in both studio art and computer science. She is a self-taught weaver with 8 years of experience. Her weaving practice explores complex textile structures that use multiple layers and the integration of conductive materials to craft electrical components within woven structures. The specific choice to use a cross-section representation came from Devendorf's experiences teaching weaving structure design in Experimental Textiles, a course at the ATLAS Institute (e.g. where cross-sections were often a more understandable format for complex drafts), and Gelosi's experiences as a student in this class. In some ways, the tool became an interactive

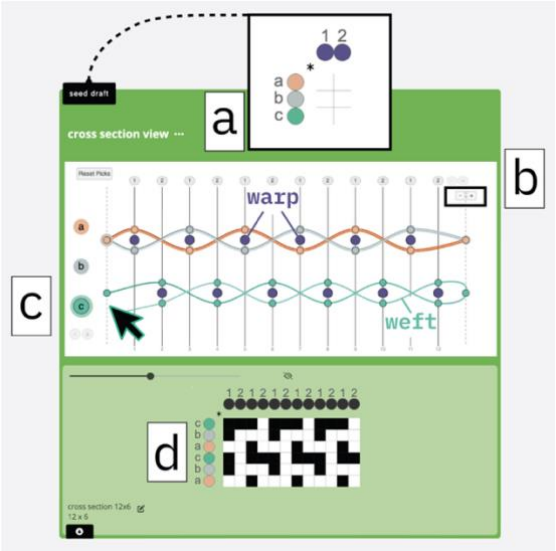


Figure 2: The cross-section operation: (a) optional seed draft defining warp/weft systems, (b) drawing canvas, (c) weft system selector, (d) the resulting draft generated in real-time.

version of the in-class worksheets Devendorf had developed for her students.

3.1. The Cross Section Operation

The cross-section tool we describe is implemented as an “operation” within AdaCAD (Devendorf et al., 2023). Operations in our software take an input draft, manipulate it according to some parameters, and then return the resulting draft. In this operation, a user can draw their cross-section, and, as they draw, see a draft that corresponds to their cross-section.

Gelosi and Devendorf co-designed the cross-section operation to embody many of the characteristics of the class worksheet within the interactivity of a digital design tool. The system relies on the description of warp and weft “systems”, which we use to mark how specific warps, or wefts, work together in a cloth. For example, a two-layer cloth might be seen to have two warp systems, and the manipulation of those systems, as groups, shape if and how layers form in the cloth. When working with multiple colors, layers are also a method of blending specific sets of warp and weft colors to achieve visual effects.

The number, and sequencing, of warp and weft systems can either be inherited from a seed draft (Fig. 2a) or manually assigned, and the number of warps — up to 24, a number we chose to match our dobby loom — are set on the canvas (Fig. 2b). By default, the operation maps warp systems to layers (e.g. warp system 1 is represented as a top layer, warp system 2 as second layer, and so on), though, individual warps can be reassigned to different layers as needed, offering flexibility when designing structures that vary the number and position of layers across the width of the cloth. To draw a cross-section, the designer selects a weft system and clicks the dots above or below each warp to trace the path of that single yarn across the cloth (Fig. 2c), much as one might describe a toolpath for a digital fabrication machine. As picks are drawn, the tool generates the corresponding draft in real-time (Fig. 2d), preserving pick order so that the cross-section's temporal sequence maps directly to the

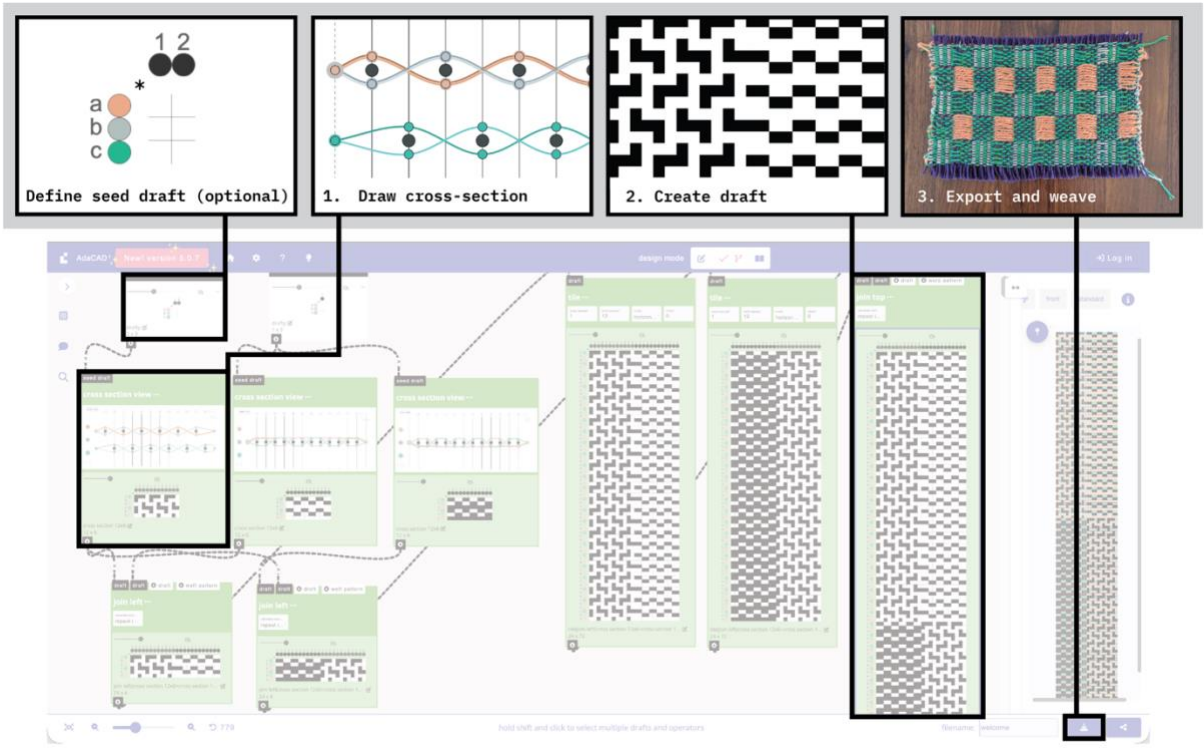


Figure 3: The four-step cross-section workflow (top) integrated into AdaCAD's broader design pipeline (bottom), where operations link to produce a weaveable draft.

production-time encoding of the draft. The cross-section can then be integrated into a larger design workflow (Fig. 3), in which operations inAdaCAD are linked together to produce a weaveable draft.

3.2. Assessment

Gelosi and Devendorf began testing the cross-sections operation through material sampling in Spring 2026 (Fig. 4). During that window, Devendorf created 11 sample designs over the course of one month (Fig. 5). This process, while less formalized, adhered to the values of autobiographical design (Desjardins & Ball, 2018; Neustaedter & Sengers, 2012). Devendorf, specifically, structured reflection by creating a journal to account for her ideas, experiences, and reflections during her use of the cross-sections tool. She labeled samples and accompanied design files with field notes about the design process and photographs of the samples produced. The insights are based on Devendorf's data; we intend them as considerations rather than generalizable design truths.

Devendorf's explorations followed two lines of inquiry: The first responded to the cross-section feature's ability to foreground color relationships in a weaving structure, the second focused on the description of complex, multilayered weaving paths. Through these inquiries Devendorf looked to unpack the kinds of questions, thinking, and forms of the experimentation that designing at the level of the cross-section illuminated for us in terms of what and how it gave "voice" to in the design process.

3.2.1 Cross-sections foregrounded the role of color and material considerations in structure design

When designing with a draft-first approach, Devendorf mentally conjures a cross-section before applying draft manipulations to get close to what she had in mind. This requires her to think in patterns and chunks–considering how weft 1 moves in relationship to weft 2 by way of draft sequence relationships. Sometimes she would have color in mind, but the complexity of these mental operations meant she would often think draft first. After drafting a structure, she would visualize it in the "viewer" of our software, which renders the design as both a 2D representation of the yarns that considers color, as well as a 3D representation.

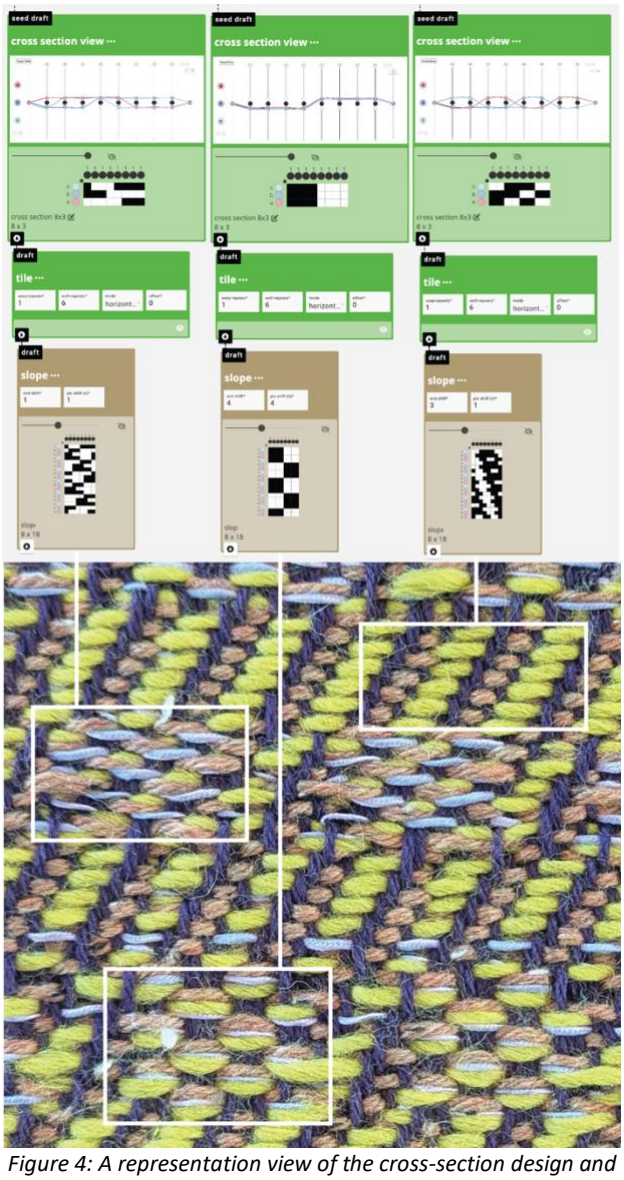


Figure 4: A representation view of the cross-section design and manipulations in AdaCAD (top) and the resulting textures of those drafts in cloth (bottom).

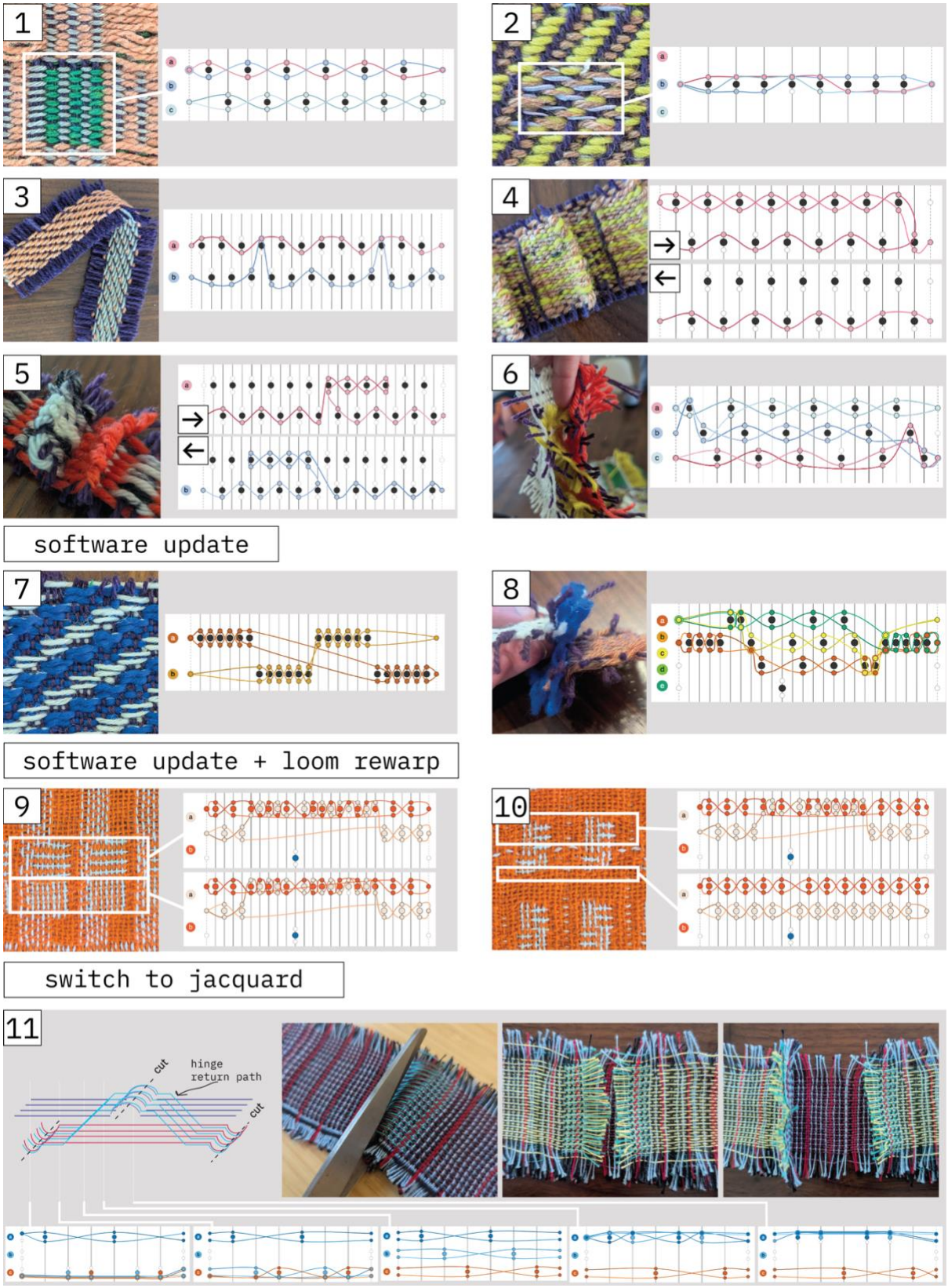


Figure 5: An overview of our sampling and testing of the cross-section tool. Each numbered entry shows a cross-section with the resulting cloth. Outlined text describes updates or events that took place and affected the cross-sections and outcome.

Cross-sections made color present immediately, prior to the start of the design. Before drawing a cross-section, Devendorf had to specify how many layers she'd like to have and how many different weft colors she planned to use. As she drew the wefts over and under the warp layers, she could clearly see what colors were intersecting, where they were on the surface, and how they would blend or chunk into different visual regions.

The cross-section tool presented the weft and warp colors and layers as components of a palette that would inform structure, rather than considerations to be decided after a structure is developed. While early iterations of the tool used a preset palette, we decided to enhance the "voice" to materials by adding the ability for the user to describe their own palettes via an input "seed" draft. The color sequences specified in the seed draft would then become present on the warps and wefts in the cross-section tool. In future iterations, we also plan to include relative material thicknesses in addition to color to support even greater variation and considerations of warp/weft variations in weaving. Even with these additions, the cross-section gives "voice" to structural and visual relationships — color, layering, and interlacement — rather than to materiality as such. Neither the cross-section nor the draft directly encodes the material properties that shape a cloth's behavior — yarn thickness, elasticity, stiffness, compressibility, or luster — which remain a matter of the designer's judgment and sampling.

3.2.2 Cross-Sections Foregrounded Weaving as Time-Sequenced Manipulation

One of the trade-offs of a cross-section representation is that it provides clarity and specificity over one dimension of the cloth, while leaving the second dimension to the imagination of the designer. Put another way, when a cross-section slices across the warps, revealing movements of wefts, it can only show a small number of wefts before the representation becomes too overwhelming and unreadable. This means that when designing cross-sections, there is an incentive to focus on simple structures that can be repeated over a small number of weft passes (e.g. tabby variations) rather than more complex repeating sequences like satin and twill. With this limitation in mind, Devendorf started using the cross-section to specify a single weft pick of each color in her weft sequence. This structure is underspecified, meaning, if she just weaved repeats of this structure, the resulting cloth would never interlace correctly and would fall apart. To manipulate these underspecified structures into valid structures, Devendorf would use our software's other operations, like "slope" and "shift," to modify the offset position of the structure along the width. This process felt more time-based and procedural than drafting, which tends to center the logic of 2D-image making. For example, she began imagining the cross-section as a kind of starting "tone" that could be sequentially modified (e.g. "pitched" or "arpeggiated") to produce specific kinds of rhythms. This, too, added an element of surprise to the sampling process. She wasn't trying to describe a specific idea, but to specify a structure and see where it might expand from there.

3.2.3 Cross-Sections Supported Detailed Attention to Interlacements in Complex Structures

One of the first potentials we saw in cross-sections was to design complex weft paths—wefts that change direction mid-way, loop, or otherwise change directions without traveling the full width of the cloth. Our collaborations with Elizabeth Meiklejohn helped us appreciate the complexity of shapes that could be achieved with these approaches, and Devendorf began her investigation of these structures by recreating several of Elizabeth Meiklejohn's samples on a 24-shaft dobby loom (Meiklejohn et al., 2024). This process helped us advance the design of the cross-section tool itself, as

well as helped us develop “hacks” for realizing complicated designs on a relatively limited number of frames. For example, we realized that we needed a way for the user to arbitrarily assign warps to different layers, allowing some regions of the design to be a single layer, and others to be two-layers. When presented together, the layer assignments of each warp also helped foreground the way that increasing the number of layers decreases the density of the warp, and helped Devendorf design interlacements that would maintain even texture across regions with varied density. Additionally, the drawing of the weft path from left to right, and from right to left, helped her consider the choreography that would be required to work the partial weft passes across her loom. For example, in our flaps sample (Fig. 5, sample 5), each flap was a flattened loop traced right to left across the warps. In the looping region, both sides of the warp were enclosed in the weft, but the base layer was interlocked on only one side, so a simple repeat would let many warps peel off the back. Seeing this clearly on the cross section (e.g. the warps only having weft on one side), make it more evident that she would need a binding or “returning” right to left pass that would interlock over the base layers and skip the flaps to ensure good structure as well as even packing.

After creating the flaps and flap variations, Devendorf attempted to translate partial weft designs into full weft designs, whereby flaps and non-flat features would be cut after weaving to release. Here, she felt that she would push the abilities of the cross-section tool by attempting a complex structural challenge involving multiple warp and weft systems. The idea was to create a “Z” hinge, where two layers of cloth could be rotated between flat and non-flat states based on the size of linkages between layers. She attempted multiple iterations, most of which failed to produce cloth that she could feasibly make into the hinge she envisioned, suggesting that the cross-section in and of itself isn’t a “silver bullet” approach for all drafting. Specifically, in the first sample, Devendorf could not “read” the cloth that was produced to know where to cut, resulting in a sample that was mistakenly cut to pieces. Next, she attempted to encode cutting those instructions into the design via clearer structural differences in cutting regions. This test resulted in a cloth that correctly indicated where to cut, and whose structure supported the envisioned function, but the limitations of our loom (e.g. the design could only use 24 ends) didn’t allow for sufficient “room” to fully bind the layers and the hinge together where we needed. She also realized that her design transferred one weft from top to bottom layer, but then never returned the weft to the top layer before the next loop, leading to a second unbound hinge that she didn’t anticipate. Next, she transferred the design to a high-density jacquard loom, where she could use more space to create strong bindings between layers. In translating the design, she separated the concept into separate regions, with separations defined by the specific behaviors of the three weft systems. She then created a cross-section for each region, before using an existing operation to repeat and stitch those regions together in larger blocks.

This example serves to illustrate that cross-sections make legible a particular kind of complexity that the draft holds implicitly. While this information can be inferred from a draft, doing so demands considerable experience; the cross-section makes visible what is happening "in between" the draft cells, where multiple warps and wefts could be crossing, or interlocking, in ways that are critical to the cloth achieving a desired structure. When expanding to the jacquard and attempting to overcome the challenge of using only simple structures in cross sections, Devendorf went back to a different operation in our software called “layer notation” that gave her better flexibility in abstracting the design into functional regions and “filling” those regions with structures that would offer particular visual and packing characteristics. This led us to see the cross-section as an intermediary between design ideas and complex cloth. We found ourselves starting sketches on paper, then in cross-sections,

and then using those cross-sections to find "key regions" of our design that we could split apart and describe with layer notation. The cross-section was a helpful steppingstone for helping us closely consider relationships between layers and interlacements that would need to be accounted for when abstracting concept to suit multi-weft structures.

4. Discussion

The findings of our first-person design exploration reveal some implications for both the design of weaving software and the documentation of complex woven structures, particularly documentation formats that can leverage interactivity. We use our discussion to suggest how different representations might work together in a weaver's toolkit. To express these implications and considerations we offer two resources: a table contrasting the relative strengths of cross-sections and drafts alongside specific reflections on key interactive design features that can increase the legibility of woven structures.

4.1. A Practice-Based Comparison of Drafts and Cross Sections

Our narratives show how the cross-section and the draft complement each other. We use our experiences to outline six key dimensions of each representation mode: purpose, scope, perspective, specificity, temporality, and learnability (Table 1). These are not absolute dichotomies; in practice we observed how the two overlap and inform one another, and how these contrasts illuminate why each representation persists and what we felt it privileges. The comparison highlights a difference in orientation: the draft tends toward the loom operator and the machine, while the cross-section tends toward the maker of the cloth. The contrast is one of emphasis — what each makes immediately legible — rather than of representational capacity.

	Draft Representation	**Cross-Section Representation**
Purpose	**Translation** to machine: encodes operational instructions for the loom	**Explanation** of structure: communicates how the cloth works to weavers
Scope	**Global**: can represent a complete repeating unit (e.g. structure) or map the entire cloth as bitmap image	**Local**: usually only shows a "moment" within a repeating design
Perspective	**Top-down, collapsed, flat**: viewed from above the loom, all layers compressed into a single plane	**Side-in, dimensional**: enhances yarn-system relationships by showing threads in cross-sectional view
Specificity	**Material-agnostic**: a single draft can be "run" with different materials, yielding different cloths from the same notation	**Material-suggestive**: color and relative size are more strongly implied, grounding the representation in a particular cloth, though material properties (e.g. thickness, elasticity) remain unencoded
Temporality	**Simultaneous**: shows all loom actions at once with no implied sequence of operations	**Sequential**: reads as a time-based narrative, showing how picks move through ends over successive iterations
Learnability	Teaches **encoding**: how to make it, i.e. the procedural logic of loom setup and operation	Teaches **structure**: what the cloth is, i.e. the physical interlacement that gives fabric its properties

4.2. Implications for Structure Design and Documentation

Most weaving textbooks and software use multiple representations, and our experience highlights how each offers different pathways for understanding the creation of woven structures. Interactive systems can leverage time-based sequencing and/or synchronized movements to bring additional perspectives to a drafting process. For instance, when someone manipulates a draft and sees a simulation change in real-time, or, in our case, when someone draws a cross section and sees how it looks as a draft, the interaction can help demystify the translation process between representations. These interactive "linked representations" help designers build intuition across levels of abstraction (Jacobs et al., 2018). What our experience suggests is that the starting, or entry point, into these interactions is also critical. Which representation users are interacting with directly brings about different relationships, ideas and possibilities for weaving. In a design context, starting from a cross-section, instead of a draft, conjured deeper considerations of color, rhythm, and “between the draft cells” interactions. In a documentation context, we might consider how representations could leverage “playability”, perhaps not just representing the finished cloth, but the sequencing of machine/material movements that will be used as the cloth comes together. We plan to consider these insights in future design interactions, for instance, by creating an operation that works in reverse (e.g. translating draft to cross-section), or further explore the potential for abstract or experimental notations to intermingle.

5. Conclusion

In sharing our experience designing and testing a cross-section first design interface for drafting, we reflect on the ways that different representations can shape a practitioner’s attention and action. We unpack the varied capacities of draft and cross-sectional representations and reflect on how interactivity might further sense-making for weavers.

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